



Indira College of Engineering & Management,
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Department of Master of Computer Applications

Academic Year – 2024-2025 Project

Title – “Cook Mate: AI-Powered Recipe Recommendation and Nutrition Analysis System.”

Group members or Individual –:

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Project Guide: - Prof. Vaishali Hatkar

Project Co-ordinator: - Prof. Shreya Shenai

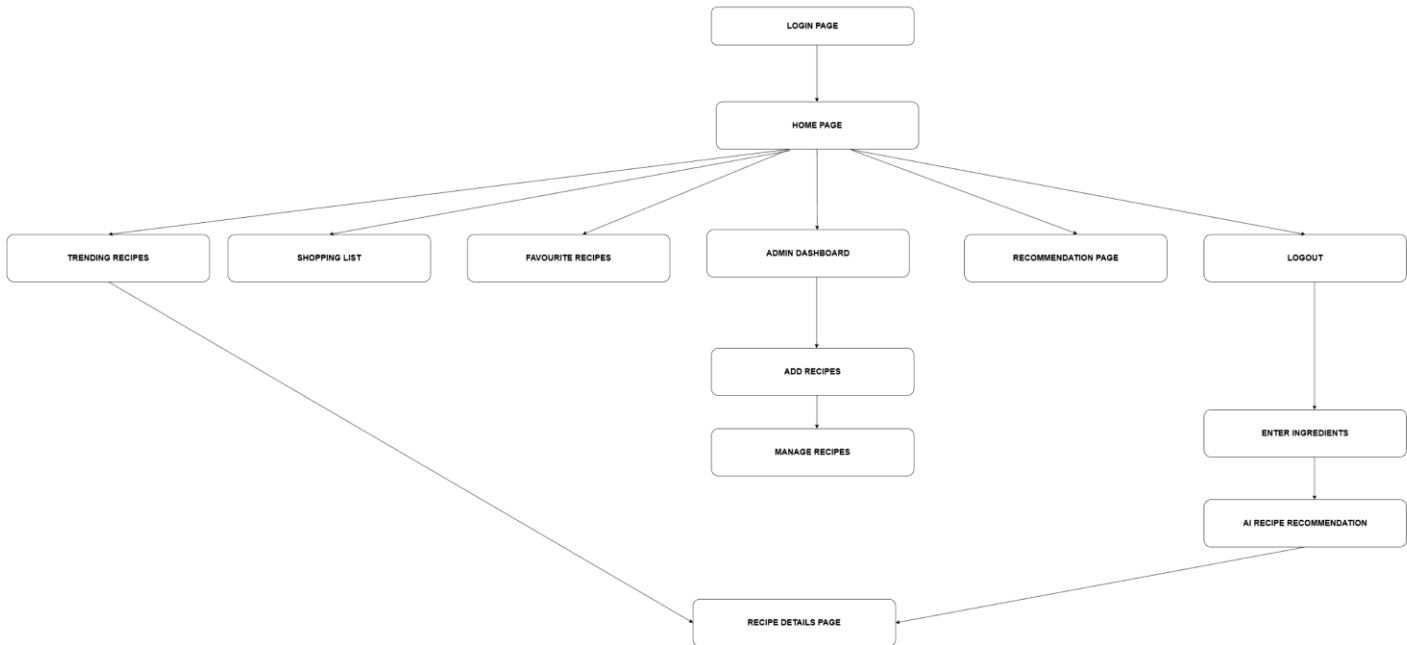
Abstract:-

Cook Mate is an AI-Powered Recipe Recommendation and Nutrition Analysis System designed to help users discover recipes based on available ingredients and health preferences. The system provides personalized recipe recommendations using ingredient matching and nutrition-based scoring techniques. It also displays nutritional information such as calories, protein, carbohydrates, and fats for each recipe. Users can manage favourite recipes, generate shopping lists for missing ingredients, and explore trending recipes through an interactive interface. The application is developed using Django, Python, HTML, CSS, and SQLite. Cook Mate aims to simplify meal planning, reduce food wastage, and promote healthier eating habits through smart food recommendation features.

Working Principle:-

The Cook Mate system works by allowing users to enter available ingredients and select their health preferences. The application processes the input and compares it with the recipe dataset stored in the database. Using ingredient matching and nutrition-based scoring logic, the system recommends the most suitable recipes to the user. Each recipe is analysed based on nutritional values such as calories, protein, carbohydrates, and fats to support healthier food choices. Users can view detailed recipe information, save recipes to favourites, and generate shopping lists for missing ingredients. The admin module manages recipes, nutritional data, and user-related content through a separate dashboard.

Project Site Map:-



Advantages:-

1. **Personalized Recommendations** – Suggests recipes based on user ingredients and preferences.
2. **Nutrition Analysis** – Displays calorie and nutrient information for healthier food choices.
3. **Food Waste Reduction** – Recommends recipes using available ingredients to minimize wastage.
4. **Easy Meal Planning** – Provides favourites and shopping list features for better organization.

Limitations:-

1. **Limited AI Intelligence** – Recommendations are based mainly on ingredient matching and scoring logic.
2. **Dataset Dependency** – System accuracy depends on the quality and completeness of the recipe dataset.
3. **Manual Nutrition Data** – Nutritional values may require manual updating for better accuracy.
4. **Limited Personalization** – The system currently does not learn user behavior dynamically over time.

Applications:-

1. **Smart Meal Planning** – Helps users plan daily meals based on available ingredients.
2. **Health and Nutrition Tracking** – Assists users in choosing healthier recipes through nutrition analysis.
3. **Food Waste Management** – Encourages efficient use of ingredients to reduce food wastage.
4. **Recipe Recommendation Platforms** – Can be used in food and cooking recommendation applications.
5. **Fitness and Diet Support** – Useful for users following specific diet or health goals.

Conclusion:-

Cook Mate is an AI-Powered Recipe Recommendation and Nutrition Analysis System that provides personalized recipe suggestions based on user ingredients and health preferences. The system combines recommendation logic with nutritional analysis to support healthier food choices and efficient meal planning. Features such as favourites, shopping lists, and recipe management improve usability and user experience. The project demonstrates the effective use of Django and recommendation system concepts in building a smart and interactive food recommendation platform.

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